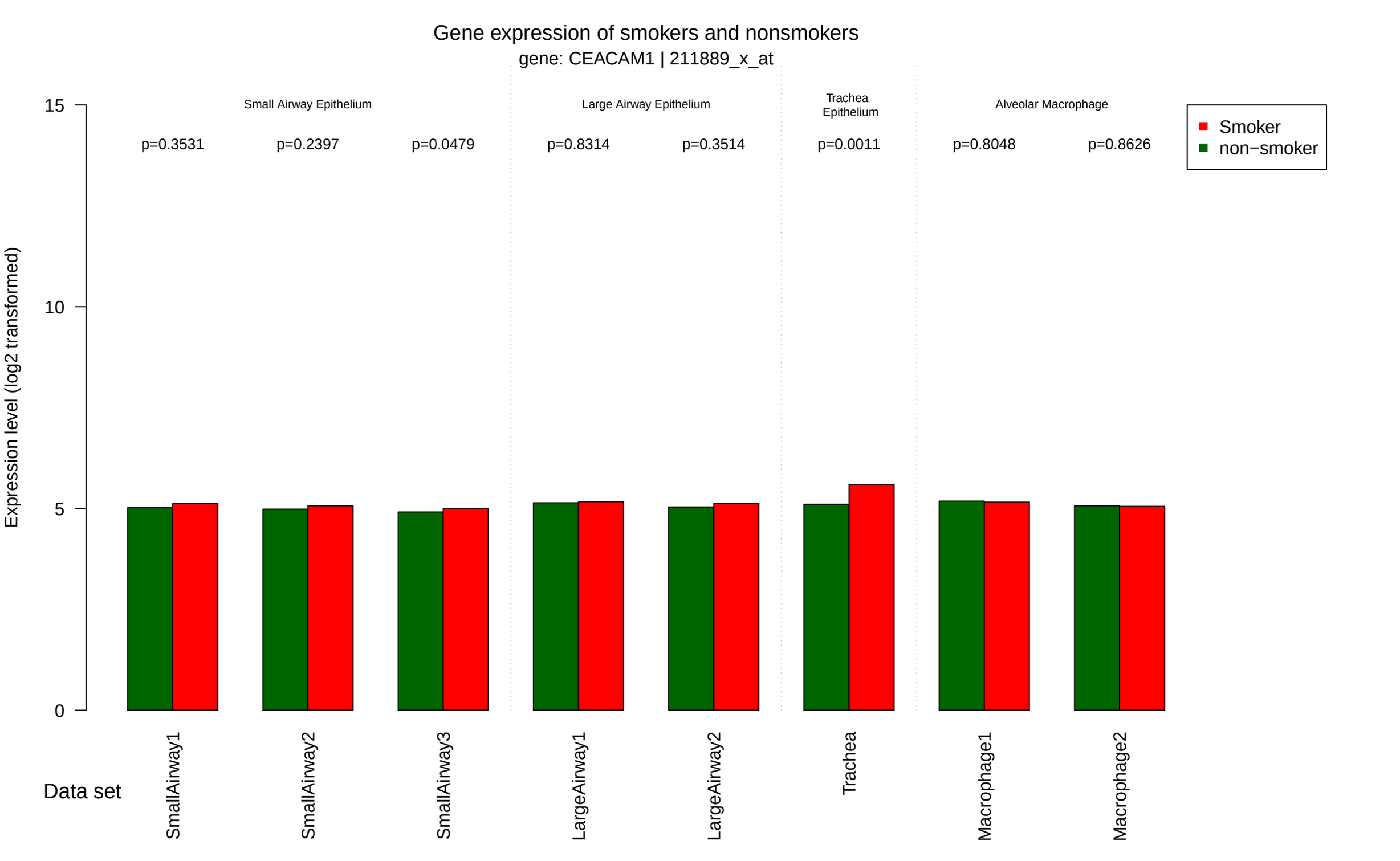
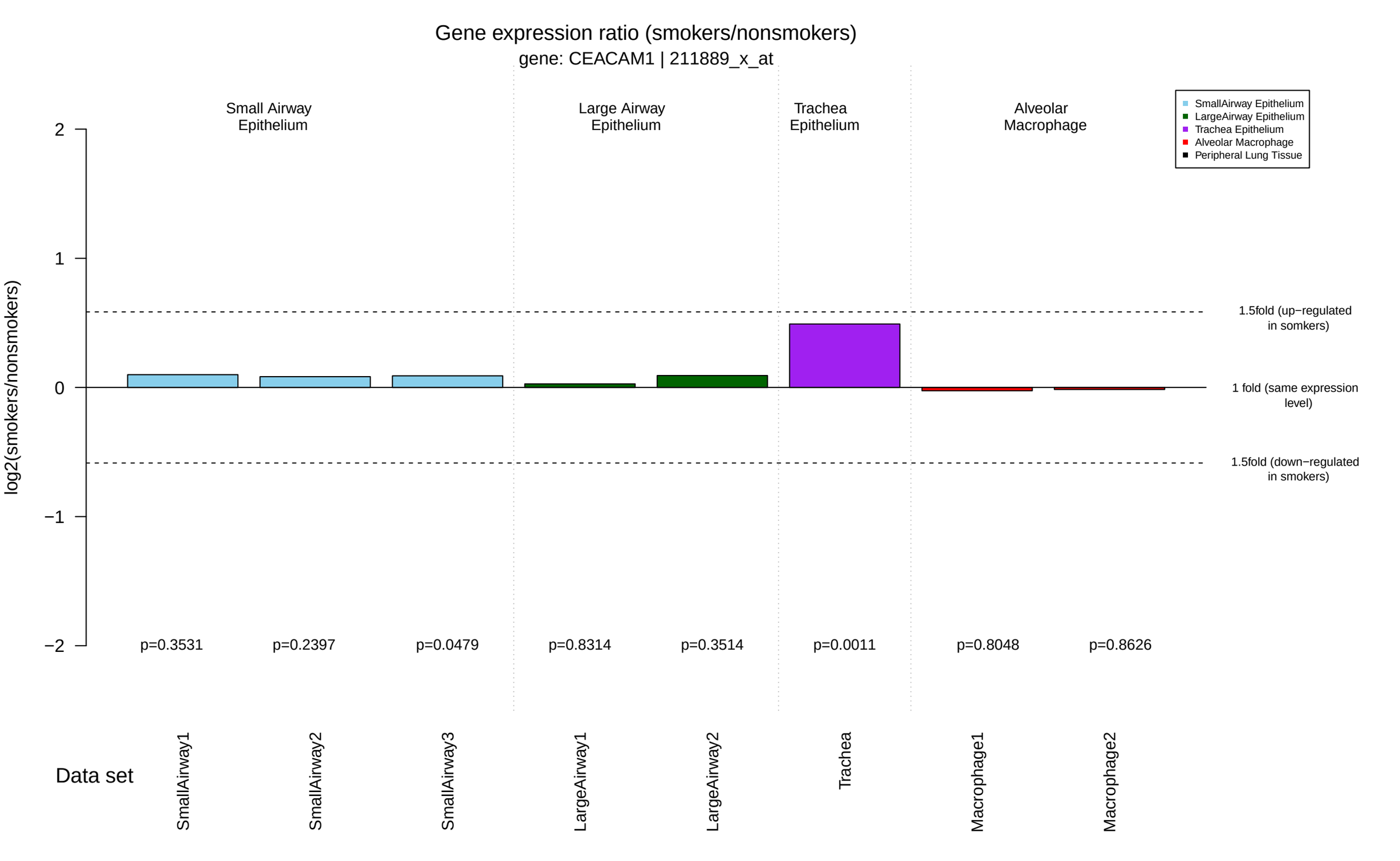
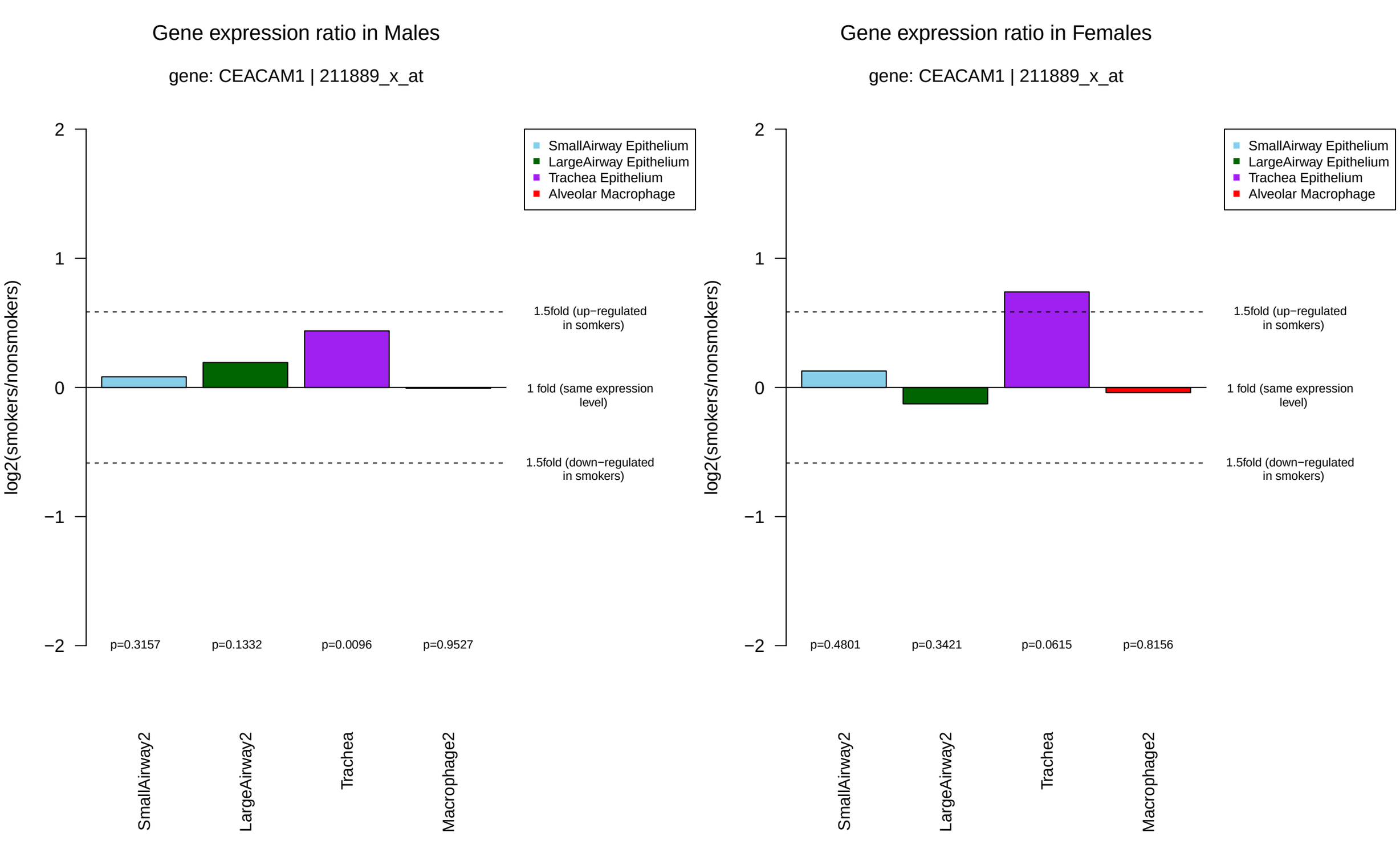
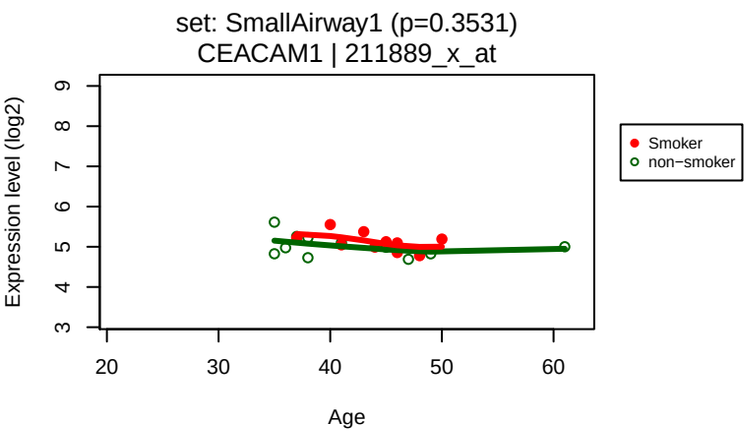
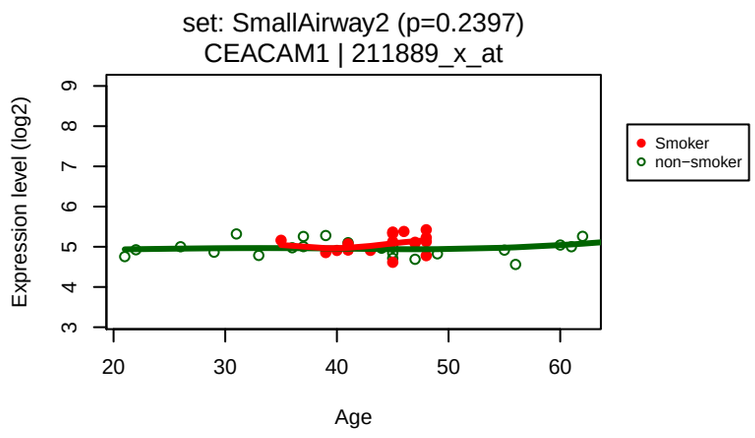
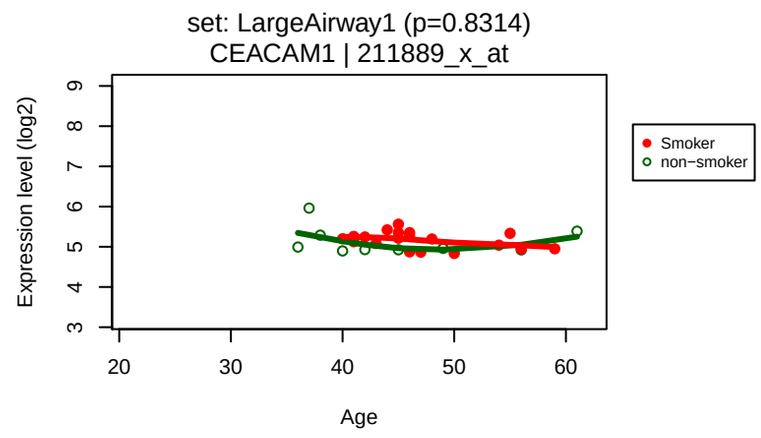

Gene expression of smokers and nonsmokers
gene: CEACAM1 | 211889_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3531
p=0.2397
p=0.0479
p=0.8314
p=0.3514
p=0.0011
p=0.8048
p=0.8626
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

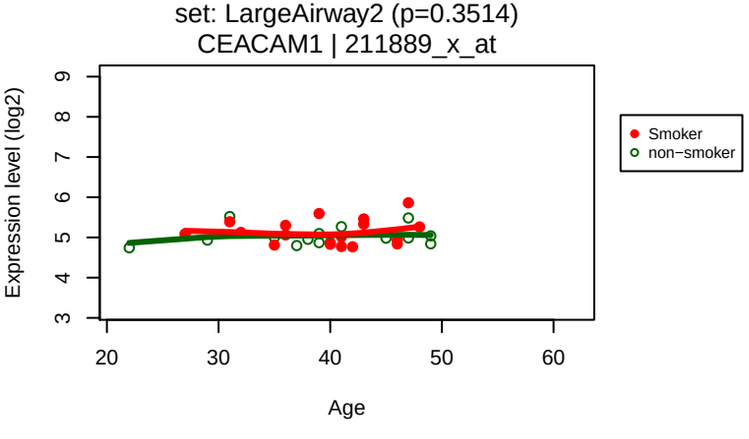

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM1 | 211889_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3531
p=0.2397
p=0.0479
p=0.8314
p=0.3514
p=0.0011
p=0.8048
p=0.8626
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

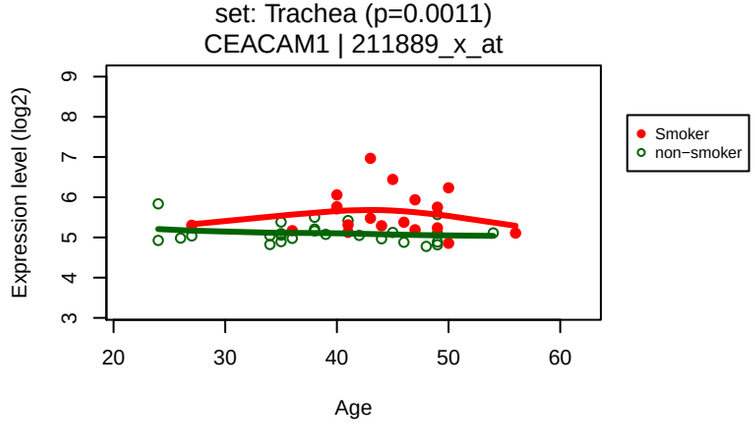

Gene expression ratio in Males
gene: CEACAM1 | 211889_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3157
p=0.1332
p=0.0096
p=0.9527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM1 | 211889_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4801
p=0.3421
p=0.0615
p=0.8156
SmallAirway2
LargeAirway2
Trachea
Macrophage2

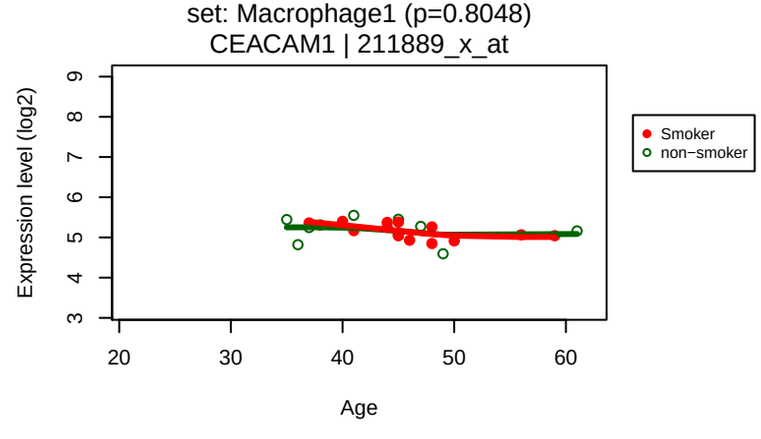

set: SmallAirway1 (p=0.3531)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker

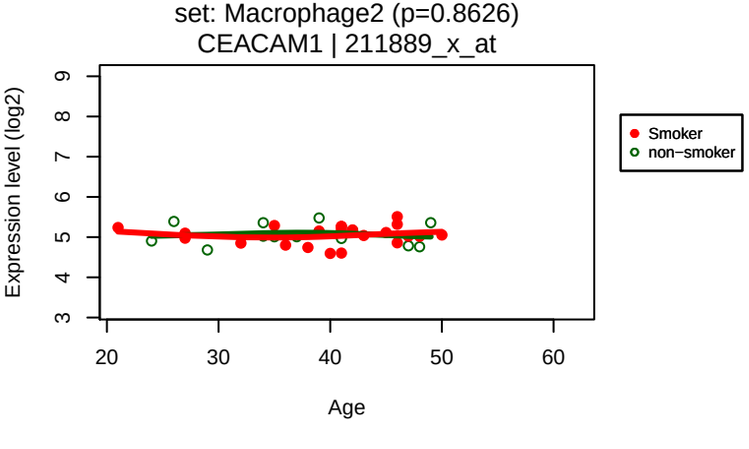

set: SmallAirway2 (p=0.2397)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker

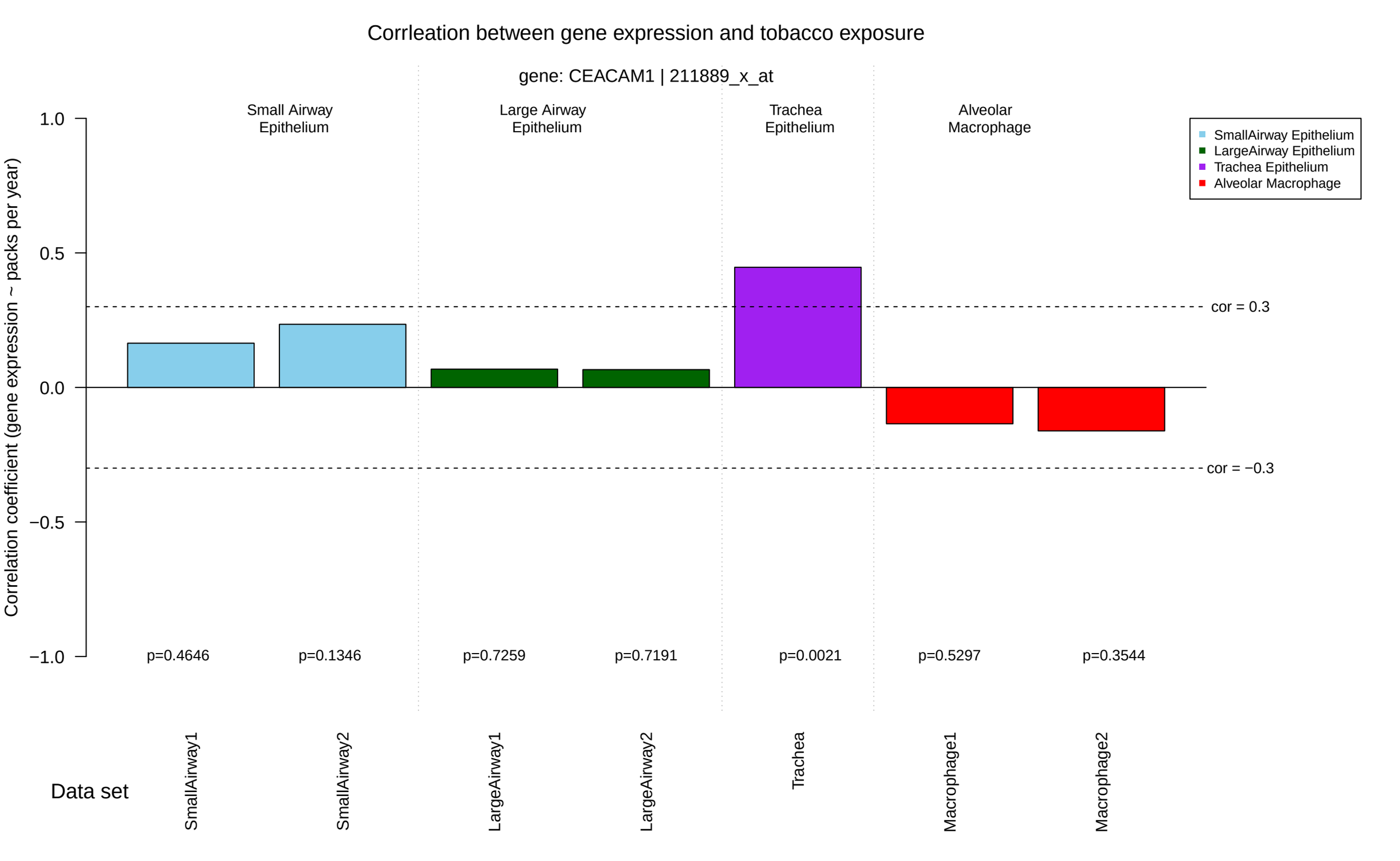

set: LargeAirway1 (p=0.8314)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3514)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8048)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8626)
CEACAM1 | 211889_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM1 | 211889_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4646
p=0.1346
p=0.7259
p=0.7191
p=0.0021
p=0.5297
p=0.3544
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2